🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 5-10 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.
Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume of Rectangular Prisms | Volume | Rectangular prism | Cubic units | Dimensions | Base area

Tap any word to see what it means and a picture.

1. The space inside a rectangular prism, found with V = l × w × h, is the ____________.

2. The amount of space inside a three-dimensional solid is its ____________.

3. A solid with six rectangular faces is a ____________.

4. The unit used to measure volume is ____________.

5. The measurements of length, width, and height of a solid are its ____________.

6 The area of the bottom face of a prism, multiplied by height to find volume, is the ______________ .

Write About the Math

How many gallons of water does the small tank hold?

¿Cuántos galones de agua tiene la pecera pequeña?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Volume of Rectangular Prisms** *Volumen de prismas rectangulares*
- ☐ **Volume** *Volumen*
- ☐ **Rectangular prism** *Prisma rectangular*
- ☐ **Cubic units** *Unidades cúbicas*
- ☐ **Dimensions** *Dimensiones*

Model: *You still multiply length × width × height even when an edge is a fraction or decimal.* — Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3$ ft^3) and that a fractional edge just means multiplying with a fraction or decimal.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The tank's volume is ____ in³ because V = ____ × ____ × ____. It holds about ____ gallons because ____ ÷ 231 ≈ ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume of Rectangular Prisms
2. **Notes 2:** Volume
3. **Notes 3:** Rectangular prism
4. **Notes 4:** Cubic units
5. **Notes 5:** Dimensions
6. **Notes 6:** Base area
7. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is 2 ft × 1.5 ft × 1 ft, a student might round 1.5 down to 1 and compute 2 × 1 × 1 = 2 ft^3, but the correct volume uses the exact edge length: 2 × 1.5 × 1 = 3 ft^3 — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
8. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — *Box A: 5 × 5 × 2 = 50 cubic in. Box B: 4 × 4 × 3 = 48 cubic in. 50 > 48, so Box A holds more.*
9. **Try It 2:** A. 18 cubic in — *V = 4 × 3 × 1.5 = 12 × 1.5 = 18 cubic in. A fractional edge does not change the steps.*